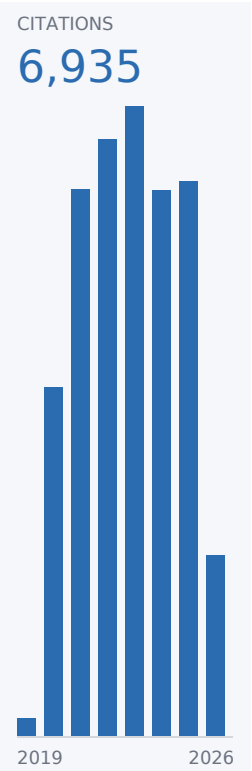


Quantum supremacy using a programmable superconducting processor

F. Arute, K. Arya, R. Babbush, D. Bacon, J. Bardin, R. Barends, R. Biswas et al.

The promise of quantum computers is that certain computational tasks might be executed exponentially faster on a quantum processor than on a classical processor1. A fundamental challenge is to build a high-fidelity processor capable of running quantum algorithms in an exponentially large computational space. Here we report the use of a processor with programmable superconducting qubits2–7 to create quantum states on 53 qubits, corresponding to a computational state-space of dimension 253 (about 1016). Measurements from repeated experiments sample the resulting probability distribution, which we verify using classical simulations. Our Sycamore processor takes about 200 seconds to sample one instance of a quantum circuit a million times—our benchmarks currently indicate that the equivalent task for a state-of-the-art classical supercomputer would take approximately 10,000 years. This dramatic increase in speed compared to all known classical algorithms is an experimental realization of quantum supremacy8–14 for this specific computational task, heralding a much-anticipated computing paradigm. Quantum supremacy is demonstrated using a programmable superconducting processor known as Sycamore, taking approximately 200 seconds to sample one instance of a quantum circuit a million times, which would take a state-of-the-art supercomputer around ten thousand years to compute.



- Quantum computer
- Quantum
- Realization (probability)
- Computer science
- Dimension (graph theory)
- Quantum algorithm

doi:10.1038/s41586-019-1666-5

Positivity preserving finite element method for the Gross–Pitaevskii ground state: discrete uniqueness and global convergence

M. Hauck, Y. Liang, D. Peterseim

We propose a positivity preserving finite element discretization for the nonlinear Gross–Pitaevskii eigenvalue problem. The method employs mass lumping techniques, which allow to transfer the uniqueness up to sign and positivity properties of the continuous ground state to the discrete setting. We further prove that every non-negative discrete excited state up to sign coincides with the discrete ground state. This allows one to identify the limit of fully discretized gradient flows, which are typically used to compute the discrete ground state, and thereby establish their global convergence. Furthermore, we perform a rigorous a priori error analysis of the proposed non-standard finite element discretization, showing optimal orders of convergence for all unknowns. Numerical experiments illustrate the theoretical results of this paper.



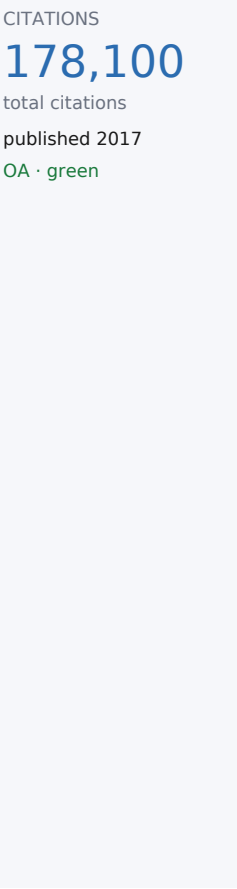
- Uniqueness
- Convergence (economics)
- Finite element method
- Element (criminal law)
- Mathematics
- Ground state

doi:10.48550/arXiv.2405.17090

Attention is All you Need

A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. Gomez et al.

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

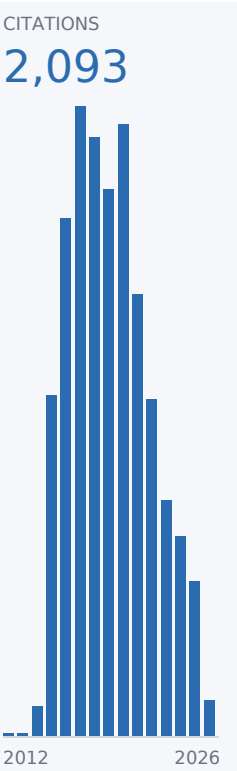


doi:10.48550/arXiv.1706.03762

Defining the role of common variation in the genomic and biological architecture of adult human height

A. Wood, T. Study, T. Esko, J. Yang, S. Vedantam, T. Pers, S. Gustafsson et al.

Timothy Frayling, Joel Hirschhorn, Peter Visscher and colleagues report a meta-analysis of genome-wide association studies for adult height in 253,288 individuals. They identify 697 variants in 423 loci significantly associated with adult height and find that these variants cluster in pathways involved in growth and together explain one-fifth of the heritability for this trait. Using genome-wide data from 253,288 individuals, we identified 697 variants at genome-wide significance that together explained one-fifth of the heritability for adult height. By testing different numbers of variants in independent studies, we show that the most strongly associated ~2,000, ~3,700 and ~9,500 SNPs explained ~21%, ~24% and ~29% of phenotypic variance. Furthermore, all common variants together captured 60% of heritability. The 697 variants clustered in 423 loci were enriched for genes, pathways and tissue types known to be involved in growth and together implicated genes and pathways not highlighted in earlier efforts, such as signaling by fibroblast growth factors, WNT/β-catenin and chondroitin sulfate-related genes. We identified several genes and pathways not previously connected with human skeletal growth, including mTOR, osteoglycin and binding of hyaluronic acid. Our results indicate a genetic architecture for human height that is characterized by a very large but finite number (thousands) of causal variants.



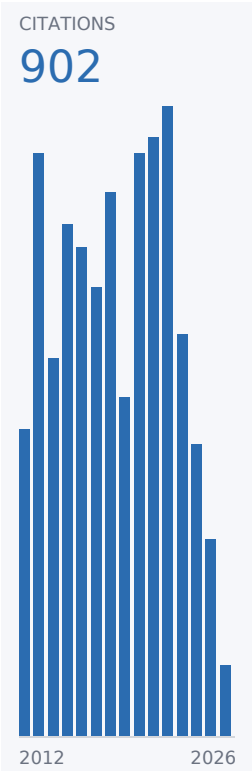
- Biology
- Genetic architecture
- Genetics
- Genome-wide association study
- Heritability
- Phenotype
- Gene

doi:10.1038/ng.3097

Sharing Detailed Research Data Is Associated with Increased Citation Rate

H. Piwowar, R. Day, D. Fridsma

BACKGROUND: Sharing research data provides benefit to the general scientific community, but the benefit is less obvious for the investigator who makes his or her data available. PRINCIPAL FINDINGS: We examined the citation history of 85 cancer microarray clinical trial publications with respect to the availability of their data. The 48% of trials with publicly available microarray data received 85% of the aggregate citations. Publicly available data was significantly (p = 0.006) associated with a 69% increase in citations, independently of journal impact factor, date of publication, and author country of origin using linear regression. SIGNIFICANCE: This correlation between publicly available data and increased literature impact may further motivate investigators to share their detailed research data.



- Citation
- Data sharing
- Clinical trial
- Data science
- Impact factor
- Aggregate data
- Computer science
- MEDLINE

doi:10.1371/journal.pone.0000308

Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set

G. Kresse, J. Furthmüller

abstract unavailable

Ab initio

Hamiltonian (control theory)

Basis set

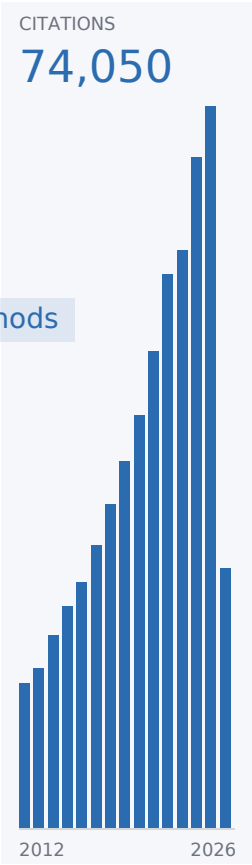
Plane wave

Statistical physics

Ab initio quantum chemistry methods

Density functional theory

Electronic structure



doi:10.1016/0927-0256(96)00008-0